

FORREST SURPRENANT

Full-Stack Software Engineer · AI Automation Architect · Forward-Deployed Engineer

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SUMMARY

Full-stack software engineer and AI automation architect with 10+ years shipping production software — and, since 2023, self-hosted agentic-AI platforms built on the Model Context Protocol (MCP). Works as a forward-deployed engineer: embedding in a business's real stack to ship autonomous multi-agent systems, LLM tool-use integrations, and workflow automation that run in production, not demos. Former defense software developer; held TS/SCI (CI polygraph) and DHS Public Trust clearances.

Core competencies: Agentic AI & Multi-Agent Systems · Model Context Protocol (MCP) · LLM Tool-Use & Integration · Workflow Automation · Full-Stack Development · Cloud/VPS Infrastructure · Forward-Deployed Delivery · Technical Consulting

TECHNICAL SKILLS

AI & Agents: Model Context Protocol (MCP) servers, autonomous multi-agent orchestration, LLM tool-use / function calling, RAG & vector search (pgvector / embeddings), OAuth 2.1 agent auth, n8n workflow automation, FastMCP / Python, prompt & context engineering, agent guardrails & evaluation

Languages: C#, TypeScript, JavaScript, Python, Java, C++, SQL (T-SQL, MySQL), PHP

Frameworks & Web: .NET Core, ASP.NET (MVC / Web Forms), Entity Framework, Angular, node.js, REST, SOAP, HTML, SASS

Cloud & DevOps: Docker, nginx, Linux (Ubuntu, Red Hat, CentOS, Kali), systemd, IIS, Apache, Git / TFS / SVN

Data & Platforms: MS SQL Server, MySQL; Sitefinity, Kentico, Sitecore, WordPress

SECURITY CLEARANCES

TS/SCI with CI Polygraph (DoD / Intelligence Community) — Inactive · DHS High-Risk Public Trust, SF-85P — Inactive

PROFESSIONAL EXPERIENCE

Managing Partner & Principal Consultant — Next Level Management Advisors LLC

Luray, VA · 2018 – Present

- Architected and self-host a fleet of 7+ Model Context Protocol (MCP) servers and 40+ production services that expose 20+ SaaS and data platforms as OAuth 2.1-secured tools LLM agents call directly.
- Built autonomous multi-agent systems that triage issues, reconcile finances, fix code, and run multichannel communications end-to-end — production automation with human-in-the-loop guardrails, not chatbots or demos.
- Operate as a forward-deployed engineer across property-tech, real-estate, and finance back-office — embedding in each client's stack to ship self-running systems (owner statements, payout reconciliation, delinquency reporting, due-diligence pipelines) with 100% client ownership of code and infrastructure.
- Direct short- and long-term property-management operations, unifying booking, banking, and ledger systems into a single source of truth.
- Advise clients on entity structuring, tax and asset protection, system automation, and cash-flow management.

Software Developer — Virginia Systems & Technology, Inc.

Warrenton, VA (Vint Hill) · Mar 2020 – Mar 2022

- Developed and enhanced software tools and capabilities in support of national-defense missions, meeting rigorous defense-industry standards.
- Collaborated with a cross-functional team to deliver high-quality solutions on schedule.
- Held a TS/SCI clearance with CI polygraph, maintaining top-tier security across all projects.

President, CEO & Treasurer — Fidum Company

Fairfax, VA · 2018 – 2022

- Led a team of 5 focused on small-business and real-estate acquisition and financing.
- Raised capital and built strategic plans that met profitability, margin, and operational targets.
- Managed both technical and business facets of client engagements and directed daily finance operations, including budgets, bank reconciliations, and financial reporting.

Senior Software Engineer — Unisys

Blue Bell, PA · Jan 2017 – Mar 2020

- Built asynchronous RESTful microservices in .NET Core and single-page applications in Angular / Material on a microservice architecture.
- Applied domain-driven design (DDD) to refactor ASP.NET MVC solutions and improve maintainability.
- Supported mission-critical systems for U.S. Customs & Border Protection (BEAGLE contract) using object-oriented design and established patterns.
- Delivered in an Agile/SCRUM environment across functional testing, performance testing, and backlog management.
- Recognized and awarded for bringing user interfaces into Section 508 accessibility compliance.

Stack: Angular 7, Material, Knockout (MVVM), TypeScript, JavaScript, jQuery, Karma, SASS, node.js, C#, LINQ, ASP.NET MVC, T-SQL

Software Engineer — Silverttech

Manchester, NH · Sep 2013 – Dec 2016

- Designed, developed, and troubleshoot high-performance, high-availability web applications in ASP.NET and C#.
- Integrated applications with external APIs including iContact, Authorize.net, and Bullhorn.
- Developed IIS extensions to manage and monitor web applications and streamline hosting.
- Built and maintained iOS and Android mobile apps on the Xamarin platform.
- Owned business analysis, development, QA, documentation, and delivery — consistently on time and under budget.

Stack: C#, LINQ, T-SQL, PHP, JavaScript, jQuery, SASS, node.js, ASP.NET Web Forms / MVC, Angular

EARLIER EXPERIENCE

- **Computer Technician / Technology Consultant** — St. John's Regional School, Concord, NH (2011 – 2013): Network manager for 200+ users on Microsoft Active Directory; built and maintained labs, printers, and routers.
- **Computer Technician** — Private Contractor, Keene, NH (2006 – 2011): Installed and supported point-of-sale systems and end-user devices for small businesses.

EDUCATION

B.S., Computer Engineering Technology — University of New Hampshire, Manchester, NH · 2015

Graduated with Honors (University Scholar). Improved speech-to-text accuracy 14% using Sphinx and Hidden Markov Models.

A.S., Computer Engineering Technology — New Hampshire Technical Institute, Concord, NH · 2013

Designed and programmed integrated circuits; programmed PLCs for robotic assembly-line arms.